

AI & Machine Learning — Task 4

Classification Models, Evaluation Metrics & Handling Imbalanced Data

Project Summary

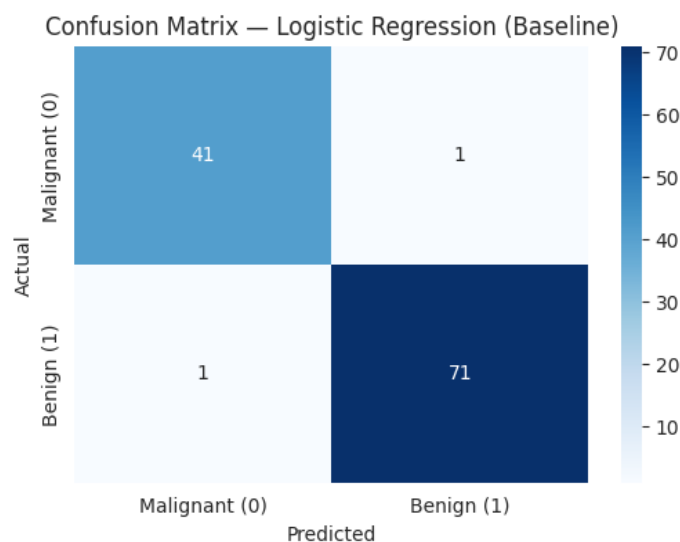
A binary classification system was built to distinguish Malignant (0) from Benign (1) tumors using the scikit-learn Breast Cancer Wisconsin dataset (569 samples, 30 numeric features). The workflow covered a stratified train/test split, feature scaling, a Logistic Regression baseline, full evaluation with confusion matrix / precision / recall / F1 / ROC-AUC, a class-imbalance experiment using class weighting, and a comparison against a Decision Tree classifier.

1. Metric Selection

Accuracy alone was not used as the primary metric. In this dataset the classes are mildly imbalanced (62.7% Benign vs 37.3% Malignant), and in a medical-diagnosis context the cost of errors is asymmetric: a False Negative (a malignant tumor predicted as benign) can delay life-saving treatment, while a False Positive (a benign tumor flagged as suspicious) only leads to a follow-up test. For this reason:

- **Recall on the Malignant class** was treated as the most business-critical metric, since it directly measures how many dangerous cases the model fails to catch.
- **F1-score** was used as a balanced summary metric per class, since it only stays high when both precision and recall are reasonably high — unlike accuracy, it cannot be inflated by simply favoring the majority class.
- **ROC-AUC** was used to judge how well the model separates the two classes across all thresholds, not just the default 0.5 cutoff.

Confusion Matrix — Baseline Logistic Regression

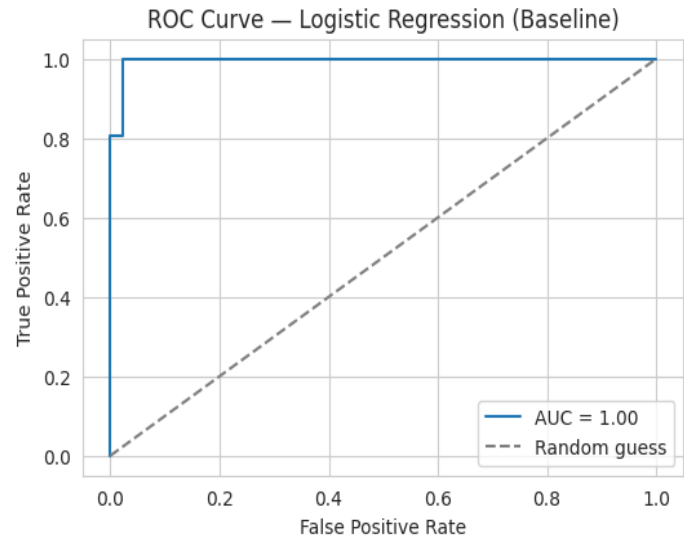


Classification Report — Baseline Logistic Regression

Class	Precision	Recall	F1-score	Support
Malignant (0)	0.98	0.98	0.98	42

Benign (1)	0.99	0.99	0.99	72
Accuracy			0.98	114

ROC Curve & AUC



The baseline model achieved an **AUC of 0.995**, indicating near-perfect separability between the two classes across all decision thresholds.

2. Imbalance Handling

The dataset shows mild class imbalance (37.3% Malignant vs 62.7% Benign), preserved identically in the train and test splits via stratified sampling. To address this, a second Logistic Regression model was trained with `class_weight="balanced"`, which re-weights the loss function inversely proportional to class frequency so that errors on the minority (Malignant) class are penalized more heavily.

Baseline vs. Balanced — Results

Model	Accuracy	F1 (Malignant)	F1 (Benign)
Logistic Regression (baseline)	98.2%	0.976	0.986
Logistic Regression (balanced)	95.6%	0.943	0.965
Decision Tree	91.2%	0.886	0.929

Finding: class weighting did not improve recall on the Malignant class in this run (it stayed at 41/42, identical to the baseline) but it increased false positives on the Benign class from 1 to 4, which lowered precision and overall accuracy (95.6% vs 98.2%) for no recall gain. This is an important, counter-intuitive result: class weighting is not automatically beneficial. It helps most when the baseline model is clearly under-predicting the minority class; here, the classes were already well separated (AUC 0.995) and the baseline handled the minority class well on its own, so re-weighting was an unnecessary trade-off for this particular dataset and split.

3. Model Comparison & Final Decision

A Decision Tree classifier was trained on the unscaled features (trees are scale-invariant) as a second model family for comparison. It reached 91.2% accuracy with F1-scores of 0.886 (Malignant) and 0.929 (Benign) — lower than both Logistic Regression variants on every metric, consistent with a single unconstrained tree overfitting the training data and generalizing less reliably than a linear model on this dataset. Decision Trees are also higher-variance: small changes in the training data can produce a noticeably different tree, whereas Logistic Regression's coefficients change more smoothly. In exchange, a Decision Tree's individual predictions are arguably easier to explain as an explicit if/then rule path, which can matter for stakeholder trust — but on pure predictive performance it was the weakest of the three models here.

Final Model Selected: Logistic Regression (baseline, no class weighting)

- Highest recall and precision on the Malignant (high-risk) class among all three models — the fewest dangerous false negatives, with no added false alarms.
- Highest ROC-AUC (0.995), showing reliable class separation across thresholds, not just at 0.5.
- Simpler and more stable than the Decision Tree, and did not need class-weighting adjustments to perform well on the minority class.

General takeaway: always verify that an imbalance-handling technique actually improves the metric that matters (here, recall on the minority/high-risk class) before adopting it by default — imbalance techniques are a tool to check, not a rule to always apply.